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Report 224

Agreement among the Government of Australia, the Government of the United Kingdom of Great Britain and Northern Ireland, and the Government of the United States of America for Cooperation Related to Naval Nuclear Propulsion

Joint Standing Committee on Treaties

November 2024

CANBERRA

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This committee is supported by staff of the Department of the House of Representatives.

Abbreviations

ACF	Australian Conservation Foundation
ADF	Australian Defence Force
ANA	Australian Nuclear Association
ANI	Australian Naval Infrastructure
AP	Protocol Additional to the Agreement between Australia and the International Atomic Energy Agency for the Application of Safeguards in Connection with the Treaty on the Non-Proliferation of Nuclear Weapons
ARPANSA	Australian Radiation Protection and Nuclear Safety Agency
ASA	Australian Submarine Agency
ASNO	Australian Safeguards and Non-Proliferation Office
CSA	Agreement between Australia and the International Atomic Energy Agency for the Application of Safeguards in Connection with the Treaty on the Non-Proliferation of Nuclear Weapons
DCCEEW	Department of Climate Change, Energy, the Environment and Water
EPA	Environment Protection Authority
EPBC Act	Environment Protection and Biodiversity Conservation Act 1999
IAEA	International Atomic Energy Agency
JSCOT	Joint Standing Committee on Treaties
NIA	National Interest Analysis
NPT	Treaty on the Non-Proliferation of Nuclear Weapons
RINA	The Royal Institution of Naval Architects
UK	United Kingdom
UNDRIP	United Nations Declaration on the Rights of Indigenous Peoples
US	United States of America

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Recommendation 1

2.35 The Committee supports the Agreement among the Government of Australia, the Government of the United Kingdom of Great Britain and Northern Ireland, and the Government of the United States of America for Cooperation Related to Naval Nuclear Propulsion and recommends that binding treaty action be taken.

Recommendation 2

3.1 The Committee recommends that the Government and the Australian Submarine Agency expand and enhance community education activities to inform the general population on how AUKUS is in the national interest, including the distinct advantages of nuclear-powered submarines.

Recommendation 3

4.22 The Committee recommends appropriate parliamentary scrutiny of any supplementary or additional instruments resulting from ratifying the AUKUS treaty.

Recommendation 4

5.26 The Committee recommends that community and worker consultation, engagement and public transparency be prioritised in the development of the plan for the management, transportation and storage of AUKUS-related nuclear waste.

Recommendation 5

5.27 The Committee recommends that advice published by the Australian Naval Nuclear Power Safety Regulator be made publicly available within a timely fashion to enhance transparency.

Recommendation 6

6.16 The Committee recommends that the Australian Government launch an AUKUS career and employment opportunities promotion campaign and further

enhance existing STEM and traditional trades engagement programs for local schools and tertiary education providers.

Recommendation 7

7.23 The Committee recommends that the Government partner with relevant State and Local governments to collaborate and coordinate on planning and delivery of enabling programs, infrastructure and works to support communities involved in the implementation of AUKUS.

Recommendation 8

7.24 The Committee recommends that the Australian Submarine Agency enhance engagement and consultation with the local Traditional Owner group and First Nations peoples; and with local environmental groups.

1. Introduction

1.1 The Joint Standing Committee on Treaties (JSCOT) was appointed by the Commonwealth Parliament to inquire into any treaty to which Australia has become a signatory, upon the treaty being tabled in the Parliament or as referred by a Minister.[1]

1.2 The Committee's resolution of appointment empowers it to inquire into and report on:

a. matters arising from treaties and related National Interest Analyses and proposed treaty actions and related Explanatory Statements presented or deemed to be presented to the Parliament;

b. any question relating to a treaty or other international instrument, whether or not negotiated to completion, referred to the committee by:

i. either House of the Parliament; or

ii. a Minister; and

c. such other matters as may be referred to the committee by the Minister for Foreign Affairs and on such conditions as the Minister may prescribe.

1.3 Prior to tabling, major treaty actions are subject to a National Interest Analysis (NIA), prepared by the Australian Government. The NIA considers the treaty, outlines the treaty obligations and any regulatory or financial implications, and reports the results of consultations undertaken with State and Territory governments, federal, state and territory agencies, and with industry or non-government organisations.

1.4 The Committee takes account of the NIA in its examination of the treaty text, in addition to other evidence taken during the inquiry.

1.5 The treaties, and matters arising from them, are evaluated to ensure ratification is in the national interest, and that unintended or negative effects on Australia do not arise.

1.6 The report contains the Committee's review of the *Agreement among the Government of Australia, the Government of the United Kingdom of Great Britain and Northern Ireland, and the Government of the United States of America for Cooperation Related to Naval Nuclear Propulsion*.

1.7 The *Agreement among the Government of Australia, the Government of the United Kingdom of Great Britain and Northern Ireland, and the Government of the United*

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[1] Joint Standing Committee on Treaties (JSCOT), 'Role of the Committee', www.aph.gov.au/Parliamentary_Business/Committees/Joint/Treaties/Role_of_the_Committee.

States of America for Cooperation Related to Naval Nuclear Propulsion establishes a legally-binding framework between Australia, the United Kingdom (UK) and the United States of America (US) to facilitate the communication and exchange of naval nuclear propulsion information between the three parties. Furthermore, the Agreement provides Australia access to critical naval nuclear propulsion information, nuclear material and equipment from both the UK and US.

1.8 A copy of the treaties considered in this report and the associated documents may be accessed through the Committee's website at: www.aph.gov.au/Parliamentary_Business/Committees/Joint/Treaties.

2. AUKUS treaty

2.1 The *Agreement among the Government of Australia, the Government of the United Kingdom of Great Britain and Northern Ireland, and the Government of the United States of America for Cooperation Related to Naval Nuclear Propulsion* was referred to the Committee on 12 August 2024.

2.2 The Committee received 260 submissions and three form letters. The Committee held four public hearings between 16 October 2024 and 24 October 2024 in Canberra, Adelaide and Perth. A list of submissions and form letters can be found at Appendix A. The hearings and witnesses are listed at Appendix B. The transcripts of evidence from the public hearings can be accessed through the Committee's website.

Overview

2.3 The purpose of the *Agreement among the Government of Australia, the Government of the United Kingdom of Great Britain and Northern Ireland, and the Government of the United States of America for Cooperation Related to Naval Nuclear Propulsion* (the Agreement) is to establish a legally-binding framework between Australia, the United Kingdom (UK) and the United States of America (US).[1] This framework will facilitate the communication and exchange of naval nuclear propulsion information between the three parties and the transfer of nuclear material and equipment from the UK and US to Australia for conventionally armed, nuclear-powered submarines.[2]

2.4 The Agreement shall remain in force until 31 December 2075.[3]

Background

2.5 On 15 September 2021, the leaders of Australia, the UK and the US announced an enhanced trilateral security partnership between the three nations entitled AUKUS.[4] The first initiative announced under AUKUS was to support Australia's acquisition of conventionally armed, nuclear-powered submarines for operation by the Royal Australian Navy.[5] On 14 March 2023, the leaders of AUKUS nations announced the

1 National Interest Analysis [2024] ATNIA 14 with attachment on consultation, *Agreement among the Government of Australia, the Government of the United Kingdom of Great Britain and Northern Ireland, and the Government of the United States of America for Cooperation Related to Naval Nuclear Propulsion* [2024] ATNIF 20 (NIA), para 3.
2 NIA, para 3.
3 NIA, para 48.
4 NIA, para 4.
5 NIA, para 4.

pathway to deliver nuclear-powered submarine capability to Australia entitled the 'Optimal Pathway'.[6]

Justifications

2.6 Australia's naval capability is reliant upon submarines as they provide a strategic advantage in terms of surveillance and protection of the nation's maritime approaches.[7] Compared to conventional submarines, the proposed conventionally armed, nuclear-powered submarines achieve superior stealth, speed, manoeuvrability, survivability, and endurance.[8] These superior submarines have greater capability to deter actions against Australia's national interests.[9]

2.7 The Agreement is critical in achieving the Optimal Pathway as it provides Australia with access to critical naval nuclear propulsion information, nuclear material and equipment from the UK and US that would not otherwise be available.[10] The Agreement provides legal authority to the UK and US to enable the transfer of naval nuclear propulsion reactors and related items to Australia, with all reactors for Australia's nuclear-powered submarine program to be provided by the UK or the US.[11] The Agreement will enable the US to transfer between three to five Virginia class submarines to Australia and will enable the UK to transfer the naval nuclear propulsion plants required for use in the SSN-AUKUS conventionally armed, nuclear-powered submarines that will be built by Australia.[12]

2.8 The provision of legal authority to transfer this nuclear information and technology that would otherwise be unauthorised under US and UK law is a 'further demonstration of AUKUS partners' commitment to implementation of the Optimal Pathway'.[13] The Agreement permits Australia to share nuclear propulsion information with authorised Australian personnel (subject to the Agreement's security obligations).[14]

2.9 By reinforcing the role of the International Atomic Energy Agency (IAEA) and its objectives throughout the lifecycle of the naval nuclear propulsion program, the Agreement 'supports Australia's commitment to the highest non-proliferation standards'.[15]

2.10 Given the sensitive nature of the nuclear technology that will be exchanged and the restrictions on the use and dissemination of nuclear information, material and equipment under US law, the Agreement includes 'robust security requirements'.[16]

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6 NIA, para 4.
7 NIA, para 5.
8 NIA, para 5.
9 NIA, para 5.
10 NIA, para 6.
11 NIA, para 6.
12 NIA, para 10.
13 NIA, para 11.
14 NIA, para 11.
15 NIA, para 9.
16 NIA, para 13.

This includes requirements on 'the use, dissemination, handling and protection of such information, material and equipment, and provides for the maintenance of mutually determined information security policies.'[17]

Obligations

Scope

2.11 Article I of the Agreement stipulates that the communication and exchange of naval nuclear propulsion information and the transfer of nuclear material and equipment relating to naval nuclear propulsion can only occur if the Originating Party determines that such cooperation will promote the objectives of AUKUS and will not constitute an unreasonable risk to the Party's defence and security.[18] Article III provides for Parties to communicate or exchange naval nuclear propulsion information and related technical data and technology necessary for the research, development or design of naval nuclear propulsion plants.[19]

2.12 Annex B of the Agreement details the nuclear material and equipment that may be transferred and stipulates that is must be transferred in complete, welded power units.[20] In addition to the equipment listed, the same Annex also lists the training, services and program supports associated with equipment that may be transferred.[21]

2.13 Article VI(A) requires cooperation conducted under the Agreement to be done so in accordance with the relevant Parties' applicable laws and regulations.[22] The Agreement recognises Australia's obligations under the Treaty on the Non-Proliferation of Nuclear Weapons (NPT), the Treaty of Rarotonga, the *Agreement between Australia and the International Atomic Energy Agency for the Application of Safeguards in Connection with the Treaty on the Non-Proliferation of Nuclear Weapons* (CSA), and the *Protocol Additional to the Agreement between Australia and the International Atomic Energy Agency for the Application of Safeguards in Connection with the Treaty on the Non-Proliferation of Nuclear Weapons* (AP).[23]

Protection of information, material and equipment

2.14 Given the sensitivity of naval nuclear propulsion information and technology, Articles VIII, IX and X of the Agreement govern its use, dissemination, handling and protection and provide for the maintenance of mutually determined information security policies.[24] Information communicated or exchanged is to be given at least the same level of protection as it is afforded by the Originating Party.[25]

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17 NIA, para 18.
18 NIA, para 14.
19 NIA, para 15.
20 NIA, para 16.
21 NIA, para 16.
22 NIA, para 17.
23 NIA, para 17.
24 NIA, para 18.
25 NIA, para 19.

2.15 Annex A stipulates the specific requirements regarding the administrative and security controls applicable to accessing and communicating this information, including requirements on physical security, personnel security, and control of the information.[26] The same Annex establishes the authority responsible for granting individual access to naval nuclear propulsion information and endorsing information security controls.[27]

2.16 Article IX of the Agreement stipulates that naval nuclear propulsion information or technology is not to be communicated to any unauthorised persons or beyond the jurisdiction or control of the Parties unless Parties give their written consent.[28]

2.17 A Memorandum of Understanding accompanying the Agreement stipulates that the UK and the US should not unreasonably withhold information, nuclear material or equipment from Australia.[29]

Nuclear non-proliferation and cooperation

2.18 All Australian nuclear-powered submarines will be conventionally armed, with the Agreement prohibiting the use of naval nuclear propulsion information, nuclear material and equipment communicated, exchanged or transferred pursuant to the Agreement for any nuclear explosive device, or for research on or development of any nuclear explosive device.[30] Australia is obligated to only use nuclear material transferred under the Agreement for naval nuclear propulsion, may not enrich or reprocess any nuclear material received, and is responsible for the management, disposition, storage and disposal of any such spent nuclear fuel and radioactive waste.[31]

2.19 Article VIII of the Agreement stipulates for all cooperation under the Agreement to require the application of Australia's CSA and AP, with all future arrangements to be negotiated by Australia with the IAEA pursuant to the Article 14 Arrangement.[32] [33] This applies to all nuclear activities within the territory of Australia, under its jurisdiction or carried out under its control anywhere in accordance with the provisions of those instruments.[34] These requirements will be implemented within Australia's existing regulatory framework and managed by the Australian Safeguards and Non-Proliferation Office (ASNO).[35]

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26 NIA, para 20.
27 NIA, para 20.
28 NIA, para 21.
29 NIA, para 40.
30 NIA, para 26.
31 NIA, para 27.
32 Article 14 Arrangement refers to the arrangement between Australia and the International Atomic Energy Agency (IAEA) that is to be negotiated and concluded pursuant to Article 14 of the Australia-IAEA Comprehensive Safeguards Agreement.
33 NIA, para 28.
34 NIA, para 28.
35 NIA, para 28.

2.20 The Agreement requires Australia to have the Article 14 Arrangement in place prior to the UK or the US transferring any nuclear material to Australia.[36] Australia is obliged to consult with the other Parties during negotiations and ensure that they share the view that the Article 14 Arrangement:

> 'protects defined categories of information, equipment and material; avoids unduly hampering, delaying or interfering with Australia's use of nuclear material and equipment; fulfils the Parties' commitment to the highest non-proliferation standard; and is consistent with the Agreement'.[37]

2.21 These requirements also apply to any proposed amendments to the Article 14 Arrangement and the Australia-IAEA Safeguards Agreements that would relate to nuclear material or equipment transferred pursuant to the Agreement.[38]

2.22 Article VII of the Agreement imposes obligations on Parties in relation to their collaboration on supporting the IAEA to implement the Australia-IAEA Safeguards Agreements and Article 14 Arrangement to meet IAEA's objectives with respect to material and equipment transferred pursuant to the Agreement.[39] This includes enabling the IAEA to verify that there is no diversion of declared nuclear material, that there is no misuse of nuclear facilities, and that there is no undeclared nuclear material or activity in Australia.[40] Parties are obliged to provide the necessary information and access that ensures their obligations are fulfilled while also protecting naval nuclear propulsion information and related classified information.[41]

Liability, indemnity and warranty

2.23 Article IV(E) requires Australia to indemnify the UK and the US against any liability, loss, costs, damage, or injury (including third party claims) arising out of, related to, or resulting from nuclear risks connected with the design, manufacture, assembly, transfer, or utilisation of any material or equipment transferred or to be transferred pursuant to the Agreement.[42] This indemnity does not apply to any conventionally armed, nuclear-powered submarine that has been in service with the US Navy until it is transferred to Australia.[43]

2.24 Article V states that the application or use of any information, nuclear material or equipment communicated, exchanged or transferred shall be the responsibility of the Recipient Party and that the other Parties do not provide any indemnity, nor do they guarantee the accuracy or completeness of the information, nor the suitability or completeness of such information, nuclear material or equipment.[44] The Agreement

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36 NIA, para 29.
37 NIA, para 29.
38 NIA, para 29.
39 NIA, para 30.
40 NIA, para 30.
41 NIA, para 31.
42 NIA, para 22.
43 NIA, para 23.
44 NIA, para 25.

provides that non-nuclear risks may be addressed by the relevant Parties in separate agreements or arrangements.[45]

Intellectual property

2.25 Article XI stipulates that where a Recipient Party (including its contractors) generates an invention that employs or is derived from information, material or equipment communicated, exchanged or transferred pursuant to the Agreement, that Party must obtain, and then transfer and assign all right, title and interest in the invention to the Originating Party.[46] The Originating Party then provides a royalty-free, non-exclusive, irrevocable, perpetual, worldwide license to the other two Parties to use that invention for government and mutual defence and security purposes.[47]

2.26 Australia is to indemnify the UK and US for any patent infringement that occurs in relation to information transferred under the Agreement.[48] The Agreement provides that Parties also waive any legal claims against other Parties for compensation, royalty or award resulting from any invention, including 'any patent application or patent, licence or sub-licence covered under relevant provisions of Article XI'.[49] Parties may establish a licensing regime for other forms of intellectual property that may arise under the Agreement.[50]

Consultation

2.27 The National Interest Analysis (NIA) states that no concerns have been identified in consultations with Commonwealth departments and State and Territory governments.[51] The NIA further states that no public consultation has been undertaken due to the Agreement's relation to national security and operational capability.[52]

Implementation and costs

2.28 Australia's implementation of the Agreement will be led by the Australian Submarine Agency, in consultation with the Department of Defence, the Department of Foreign Affairs and Trade, ASNO, the Attorney-General's Department, and other relevant Commonwealth departments and agencies.[53] No legislation or amendments to existing legislation are required to implement the Agreement.[54] However, Professor Donald Rothwell noted at the first public hearing for the inquiry that the Agreement references several existing treaties which will intersect with the Agreement, and the

45 NIA, para 24.
46 NIA, para 33.
47 NIA, para 33.
48 NIA, para 34.
49 NIA, para 34.
50 NIA, para 35.
51 NIA, paras 53-54.
52 NIA, para 55.
53 NIA, para 42.
54 NIA, para 44.

Agreement itself will 'become a foundation for additional instruments that will…need to be concluded over the life of the Agreement to give effect to it'.[55]

2.29 Any costs that may arise in relation to the transfer of nuclear material, naval nuclear propulsion plants or related equipment will be negotiated and set out in either implementing instruments or in other government-to-government mechanisms.[56]

2.30 The Agreement shall remain in force until 31 December 2075.[57]

Withdrawal

2.31 Parties are able to terminate the Agreement by providing at least one year's written notice to the other Parties.[58] A Party considering a determination of breach or withdrawal is expected to consult with the other Parties prior to finalising its decision.[59] If a Party terminates or breaches the Agreement, each other Party has the right to require the return or destruction of any naval nuclear propulsion information, nuclear material and equipment that it communicated, exchanged, or transferred.[60]

2.32 If Australia breaches its obligations imposed by, terminates, abrogates, or otherwise withdraws from the NPT, CSA or Article 14 Arrangement, or detonates a nuclear explosive device, the other Parties have the right to cease further cooperation under the Agreement and require the return of any nuclear material or equipment transferred pursuant to the Agreement.[61]

2.33 Any disagreement that may arise between the Parties in the implementation or interpretation of the Agreement must be resolved through mutual consultations without recourse to any other dispute settlement mechanisms.[62]

Future treaty action

2.34 The Agreement may be amended if Parties agree to do so in writing and shall enter into force on the date of the last note in an exchange of diplomatic notes among the Parties.[63]

Recommendation 1

2.35 The Committee supports the *Agreement among the Government of Australia, the Government of the United Kingdom of Great Britain and Northern Ireland,*

55 Professor Donald Rothwell, Private Capacity, *Committee Hansard*, Canberra, 16 October 2024, p. 5.
56 NIA, para 45.
57 NIA, para 48.
58 NIA, para 48.
59 NIA, para 50.
60 NIA, para 49.
61 NIA, para 51.
62 NIA, para 36.
63 NIA, para 47.

and the Government of the United States of America for Cooperation Related to Naval Nuclear Propulsion **and recommends that binding treaty action be taken.**

3. Benefits of nuclear powered submarines

Overall benefits

3.1 Submarines have long been a critical part of Australia's defence strategy.[1] The Royal Australian Navy currently possesses six diesel electric Collins class submarines which came into service in the 1990s and were initially meant to be coming to the end of their useful lives in the 2020s. However, a transition program is currently in place to extend their use until the submarines acquired under the AUKUS program are in operation.[2]

3.2 The Committee heard from the Australian Submarine Agency (ASA) that 'Australia is facing the most complex and challenging strategic circumstances in our region since the Second World War', circumstances which led the Australian Government to commit to acquiring nuclear-powered, conventionally armed submarines to refresh its fleet.[3]

3.3 Nuclear-powered submarines have a number of benefits over conventionally powered submarines, including being faster and more manoeuvrable.[4] One of the biggest benefits of nuclear power is that the submarines do not have to surface to recharge on fuel, as is the case for the Collins class, greatly extending their range capabilities.[5] Instead, the range of nuclear-powered submarines is only restricted by the need to restock provisions for the crew. Nuclear-powered submarines also offer superior amenities for crew members, which further increases the time a vessel can stay submerged.[6]

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1 The Australia Institute, *Submission 230, Attachment 1*, p. 14.

2 Ms Alexandra Kelton, Head of Policy, Strategy and Engagement Division, Australian Submarine Agency (ASA), *Committee Hansard*, Canberra, 24 October 2024, p. 9.

3 Ms Alexandra Kelton, Head of Policy, Strategy and Engagement Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, p. 6.

4 National Interest Analysis [2024] ATNIA 14 with attachment on consultation, *Agreement among the Government of Australia, the Government of the United Kingdom of Great Britain and Northern Ireland, and the Government of the United States of America for Cooperation Related to Naval Nuclear Propulsion* [2024] ATNIF 20 (NIA), para 5; Ms Alexandra Kelton, Head of Policy, Strategy and Engagement Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, p. 6; Commodore Michael Jacobson, Director General, Submarines, Department of Defence, *Committee Hansard*, Canberra, 24 October 2024, p. 12.

5 Mr Robin Gehling, Secretary, Australian Division, The Royal Institution of Naval Architects (RINA), *Committee Hansard*, Canberra, 24 October 2024, p. 1.

6 Mr Robin Gehling, Secretary, Australian Division, RINA, *Committee Hansard*, Canberra, 24 October 2024, p. 2.

3.4 As explained by Commodore Michael Jacobson from the Department of Defence at a public hearing for the inquiry:

> 'As far to the utility of a nuclear-powered submarine, it's a combination of stealth, range, endurance and its ability to operate undetected. It can choose where it operates and when in a way that a conventional, or diesel powered, submarine can't. That nuclear-powered submarine will be able to maintain our requirements into the future, where the relative capability of our current diesel submarines will start to wane into the decades to come.'[7]

3.5 The Committee did receive conflicting evidence on the stealth of a nuclear-powered submarine versus a conventionally powered one. The Australia Institute stated that 'conventionally powered submarines can be significantly quieter than their nuclear-powered counterparts',[8] however acknowledged that a nuclear-powered submarine whose location was compromised would be able to get away easier.[9] In contrast, the ASA maintains that nuclear-powered submarines are stealthier than their conventional counterparts.[10]

AUKUS partner's experience

3.6 Although nuclear-powered submarines are an established technology in several other countries,[11] it is acknowledged that the Australian Navy will require the assistance of its AUKUS partners to learn how to operate and maintain them. This is due to the state of nuclear science not being as developed here in Australia as it is in the United Kingdom (UK) and the United States of America (US).[12]

3.7 While Australia does have some nuclear industry,[13] the extra challenges presented by designing and building nuclear-powered submarines will require an advanced level of expertise and experience not currently available in Australia.[14]

3.8 Mr Robin Gehling, Australian Division Secretary of the Royal Institution of Naval Architects explained:

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7 Commodore Michael Jacobson, Director General, Submarines, Department of Defence, *Committee Hansard*, Canberra, 24 October 2024, p. 12.

8 The Australia Institute, *Submission 230, Attachment 2*, p. 19.

9 The Australia Institute, *Submission 230, Attachment 1*, p. 17.

10 Ms Alexandra Kelton, Head of Policy, Strategy and Engagement Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, p. 5.

11 Dr John Harries, Secretary, Australian Nuclear Association (ANA), *Committee Hansard*, Canberra, 16 October 2024, p. 1.

12 Answers to questions on notice provided by the ASA at a public hearing in Canberra, 24 October 2024, p. 1.

13 Dr John Harries, Secretary, ANA, *Committee Hansard*, Canberra, 16 October 2024, p. 2.

14 Dr John Harries, Secretary, ANA, *Committee Hansard*, Canberra, 16 October 2024, p. 2; Mr Robin Gehling, Secretary, Australian Division, RINA, *Committee Hansard*, Canberra, 24 October 2024, p. 2.

> 'It's been 30 years since the Collins class was built, and, of course, nuclear-propelled submarines are another step further advanced than the conventional Collins class. The skills gap needs to be addressed right through the project.'[15]

3.9 It is for this reason that the cooperation of the US and UK in training Australian personnel is instrumental to Australia gaining sovereign nuclear-powered submarine capacity. This is being achieved by Australian personnel traveling overseas to train in the US and UK, as well as the facilitation of US and UK submarines traveling to Australia for training opportunities.

3.10 The Committee heard evidence of the following training cooperation between the AUKUS members:

- US laws have been changed to enable Australians to work in US shipyards and for maintenance to be done on US submarines in Australia;[16]
- 40 people from ASC Pty Ltd. are working in the Pearl Harbour Naval Shipyard;[17]
- Preparations are underway for US Virginia class and UK Astute class submarines to be docked at HMAS *Stirling* from 2027;[18]
- 260 Australians have received 'nuclear-specific training and familiarisation briefings';[19]
- 60 Royal Australian Navy personnel are currently training in the US, with a further 100 starting in 2025;[20]
- Six officers have finished their training and are working on Virginia class submarines;[21]
- More Australian sailors who have finished their studies are expected to join US submarine crews by the end of 2024;[22]
- In the UK, three Royal Australian Navy officers have finished a UK nuclear reactor course and are now working on UK Astute class submarines, with more officers expected to begin that training later in 2024;[23]

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15 Mr Robin Gehling, Secretary, Australian Division, RINA, *Committee Hansard*, Canberra, 24 October 2024, p. 1.

16 Ms Alexandra Kelton, Head of Policy, Strategy and Engagement Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, p. 11.

17 Ms Alexandra Kelton, Head of Policy, Strategy and Engagement Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, p. 13.

18 Ms Alexandra Kelton, Head of Policy, Strategy and Engagement Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, p. 11.

19 Ms Alexandra Kelton, Head of Policy, Strategy and Engagement Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, p. 6.

20 Ms Alexandra Kelton, Head of Policy, Strategy and Engagement Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, pages 12-13.

21 Ms Alexandra Kelton, Head of Policy, Strategy and Engagement Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, p. 12.

22 Ms Alexandra Kelton, Head of Policy, Strategy and Engagement Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, pages 12-13.

23 Ms Alexandra Kelton, Head of Policy, Strategy and Engagement Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, p. 13.

- The UK's Royal Navy work with the ASA by delivering naval nuclear propulsion training for more than 250 Australian personnel;[24]
- Australians are embedded in program-delivery teams in the UK Ministry of Defence.[25]

3.11 These initiatives further the Agreement's intention to 'support Australia's development of the infrastructure, technical capabilities, industry, and human capital necessary to produce, maintain, operate, and steward a sovereign fleet of conventionally armed, nuclear-powered submarines'.[26]

Conventionally armed

3.12 Several witnesses presented evidence to the Committee with concerns that the radioactive material contained in submarine reactor units could be weaponised, due to the belief that it would be 'weapons grade' uranium.[27]

3.13 The Agreement articulates that the submarines acquired by Australia will be conventionally armed,[28] and that Australia is not seeking nuclear weapons:

> The Agreement obligates Australia to only use nuclear material transferred under the Agreement for naval nuclear propulsion, to not enrich or reprocess any nuclear material received.[29]

3.14 Furthermore, the nuclear reactors located within the submarines will be powered by a different type of nuclear technology than that required to create nuclear weapons.[30]

3.15 Dr Craig Everton from the Australian Safeguards and Non-Proliferation Office explained that 'weapons grade' uranium does not have an internationally standard definition and is not a term used by the International Atomic Energy Agency (IAEA), who instead use the terms 'low-enriched' and 'high-enriched'. As explained by Dr Everton:

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24 Ms Alexandra Kelton, Head of Policy, Strategy and Engagement Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, p. 13.

25 Ms Alexandra Kelton, Head of Policy, Strategy and Engagement Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, p. 13.

26 Agreement among the Government of Australia, the Government of the United Kingdom of Great Britain and Northern Ireland, and the Government of the United States of America for Cooperation Related to Naval Nuclear Propulsion, p. 2.

27 Mr David Sweeney, Nuclear Policy Analyst, Australian Conservation Foundation, *Committee Hansard*, Canberra, 16 October 2024, p. 19; Friends of the Earth Adelaide, *Submission 5*, p. 3; Caring for South Australia, *Submission 159*, p. 3.

28 Agreement among the Government of Australia, the Government of the United Kingdom of Great Britain and Northern Ireland, and the Government of the United States of America for Cooperation Related to Naval Nuclear Propulsion, Article IV (A).

29 NIA, para 27.

30 Dr John Harries, Secretary, ANA, *Committee Hansard*, Canberra, 16 October 2024, p. 3.

> The IAEA…categorised nuclear material into source material and special fissionable material and then subcategorises that between less than 20 per cent, which is low-enriched, and more than 20 per cent, which is high-enriched.[31]

3.16 While the uranium used in the submarine reactors will be high-enriched, it will not be in a form that could be used to create weapons. To do so would require further chemical processing in facilities that Australia 'doesn't have and will not seek out', according to the ASA.[32]

3.17 The Agreement also includes the following specifications prohibiting the use of information, material or equipment provided under the Agreement from being used in pursuit of nuclear weapons:

> Information, Material, and Equipment communicated, exchanged, or transferred pursuant to this Agreement shall not be used for any nuclear explosive device, or for research on or development of any nuclear explosive device.[33]
>
> Information (including information incorporated in a production or utilization facility or important component part thereof) that is important to the design, construction, fabrication, operation, or maintenance of a uranium enrichment or nuclear fuel reprocessing facility or a facility for the production of heavy water shall not be communicated or exchanged pursuant to this Agreement.[34] [35]

Committee comment

3.18 The Committee was interested to hear of the distinct advantages of the nuclear-powered submarines in contrast to the existing Collins class vessels.

3.19 While acknowledging public concern that the material used to power the submarine reactors will highly enriched, the Committee was pleased to hear that it will not be in a form that could be used to create weapons.

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31 Dr Craig Everton, Assistant Secretary, Non-Proliferation, Technology and Partnerships, Australian Safeguards and Non-Proliferation Office, Department of Foreign Affairs and Trade, *Committee Hansard*, Canberra, 24 October 2024, p. 7.

32 Ms Sophia Blix, Acting Head, Stewardship and Security Division, Australian Submarine Agency, *Committee Hansard*, Canberra, 24 October 2024, pages 7-8.

33 Agreement among the Government of Australia, the Government of the United Kingdom of Great Britain and Northern Ireland, and the Government of the United States of America for Cooperation Related to Naval Nuclear Propulsion, Article VI (C).

34 Heavy water is used in nuclear reactors.

35 Agreement among the Government of Australia, the Government of the United Kingdom of Great Britain and Northern Ireland, and the Government of the United States of America for Cooperation Related to Naval Nuclear Propulsion, Article VI (F).

Recommendation 2

3.1 The Committee recommends that the Government and the Australian Submarine Agency expand and enhance community education activities to inform the general population on how AUKUS is in the national interest, including the distinct advantages of nuclear-powered submarines.

4. Wording of the Agreement

Withdrawal from the Agreement

4.1 In submissions to the inquiry and at public hearings, the Committee was presented with evidence that raised concerns with the potential imbalance of the Agreement.

4.2 At the first public hearing in Canberra, Professor Donald Rothwell stated that the provisions concerning withdrawal from the Agreement offered 'an enormous amount of flexibility' to the United Kingdom (UK) and the United States of America (US).[1]

4.3 Professor Rothwell's submission cited the term of 'Originating Party' in Article I as one such example, in which an 'Originating Party' is able to withdraw from the Agreement if they cannot determine that cooperation under the Agreement 'will promote and will not constitute an unreasonable risk to its defence and security'.[2] [3] Professor Rothwell highlighted that under the terms of Article II(N) of the Agreement, the definition of an 'Originating Party' only extends to the UK or US.[4]

4.4 Professor Rothwell further described the Agreement as 'somewhat exceptional' as many other international treaties provide a more limited or constrained set of circumstances under which a party may withdraw.[5] He theorised that this flexibility could be a result of the long lifespan of the Agreement as well as the lack of precedent for the UK and US entering into such agreements.[6]

4.5 The Australia Institute also proposed rewording or adding provisions that afforded Australia more flexibility in withdrawing from the Agreement.[7] Mr Allan Behm suggested that Australia should take precautions and not sign away 'the potential for future governments to take decisions in the national interest, whether they're in the national security interest or… the national economic interest'.[8]

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1 Professor Donald Rothwell, Private Capacity, *Committee Hansard*, Canberra, 16 October 2024, p. 6.
2 Professor Donald Rothwell, Private Capacity, *Submission 138*, p. 5.
3 Agreement among the Government of Australia, the Government of the United Kingdom of Great Britain and Northern Ireland, and the Government of the United States of America for Cooperation Related to Naval Nuclear Propulsion, Article I.
4 Professor Donald Rothwell, Private Capacity, *Submission 138*, p. 5.
5 Professor Donald Rothwell, Private Capacity, *Committee Hansard*, Canberra, 16 October 2024, p. 6.
6 Professor Donald Rothwell, Private Capacity, *Committee Hansard*, Canberra, 16 October 2024, p. 6.
7 Mr Allan Behm, Director, International and Security Affairs, The Australia Institute, *Committee Hansard*, Canberra, 16 October 2024, p. 12.
8 Mr Allan Behm, Director, International and Security Affairs, The Australia Institute, *Committee Hansard*, Canberra, 16 October 2024, p. 12.

Indemnity and dispute resolution

4.6 The Committee heard concerns relating to indemnity provisions found in the Agreement. Professor Rothwell highlighted that Australia was acquiring all the risk relating to the nuclear-powered submarines built under the Agreement, with no risk or liability being placed upon the UK or US.[9] Rothwell said that as the Agreement stands, Australia gains the legal responsibility for the submarines and the UK and US are indemnified for any issues that may have arisen throughout the build process.[10]

4.7 The Australian Conservation Foundation (ACF) shared these concerns and added that the 'broad' indemnity provisions exposed Australia to significant financial risk and were not in the national interest.[11] The ACF suggested that the Committee take legal advice on the scope of the indemnity to map out in detail the potential liability that the Agreement placed on Australia.[12]

4.8 The Australian Submarine Agency (ASA) stated that Australia indemnifying the UK and US for liability arising from nuclear risk was 'consistent with Australia's commitment to being a sovereign and responsible steward of naval nuclear propulsion technology that will be owned and operated by Australia as a sovereign capability'.[13]

4.9 When appearing before the Committee, Professor Rothwell stated that an 'unusual' aspect of the Agreement was that it did not contain 'some of the consistent formal mechanisms for resolution of disputes or disagreements that one might otherwise find in a treaty-level instrument dealing with matters of such complexity and significance'.[14] Rothwell said that modern treaties with the level of detail contained in the Agreement typically contained a provision for arbitration if a dispute arose between parties and that the complexity of the arrangements contained within the Agreement created the potential for 'fairly high-level international disagreements'.[15]

Tripartite commitments and intersection with other international instruments

4.10 The importance of the Agreement in facilitating the AUKUS program was explored by the Committee. Professor Rothwell stated that it would not be possible for Australia to acquire nuclear-powered submarines through AUKUS without a treaty-level instrument of this type being ratified.[16]

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9 Professor Donald Rothwell, Private Capacity, *Committee Hansard*, Canberra, 16 October 2024, p. 7.
10 Professor Donald Rothwell, Private Capacity, *Committee Hansard*, Canberra, 16 October 2024, p. 7.
11 Australian Conservation Foundation (ACF), *Submission 181,* p. 5.
12 Mr Adam Beeson, General Counsel, ACF, *Committee Hansard*, Canberra, 16 October 2024, p. 21.
13 Ms Alexandra Kelton, Head of Policy, Strategy and Engagement Division, Australian Submarine Agency (ASA), *Committee Hansard*, Canberra, 24 October 2024, p. 6.
14 Professor Donald Rothwell, Private Capacity, *Committee Hansard*, Canberra, 16 October 2024, p. 7.
15 Professor Donald Rothwell, Private Capacity, *Committee Hansard*, Canberra, 16 October 2024, p. 7.
16 Professor Donald Rothwell, Private Capacity, *Committee Hansard*, Canberra, 16 October 2024, p. 5.

4.11 The ASA stated that the length of the Agreement was a demonstration of the longevity of the partnership between the three Parties.[17] ASA said the three Parties were committed to AUKUS and that the Agreement made the necessary commitments to 'collectively uplift the industrial capacity and trilateral defence of all three countries'.[18]

4.12 ASA cited US Congress passing legislation to allow the sale of Virginia Class submarines to Australia, investments in Australian industrial bases, and the authorisation of Australian workers on US shipyards as examples of the commitment from Australia's AUKUS partners.[19]

4.13 The Agreement's relationship with other international agreements and instruments was discussed. Rothwell described the Agreement's preamble referencing of six other existing agreements, protocols and treaties as 'exceptional' and a reinforcement of the importance of the international law framework.[20]

4.14 Rothwell stated that these intersecting relationships in tandem with the lifespan of the Agreement would result in the Agreement becoming a 'foundation for additional instruments' so that it could remain fit for purpose.[21] The ASA agreed with this assessment, stating that while the purpose of the Agreement was specific, the scope of the initiative was large and legally complex.[22] The ASA stated that additional agreements will be entered into to facilitate the purpose of the Agreement.[23] Professor Rothwell suggested that the Committee could recommend that there is appropriate parliamentary scrutiny of any instruments created as a result of the Agreement.[24]

Nuclear weapons

4.15 The potential for the Agreement to create a 'slippery slope' towards nuclear weapons was repeatedly raised in submissions and at public hearings. The ACF raised this concern and stated that Australia should sign the Treaty on the Prohibition of Nuclear Weapons before ratifying the Agreement to avoid Australia developing sovereign nuclear weapons capability or hosting UK and US nuclear forces.[25] The ACF stated that signing the Treaty on the Prohibition of Nuclear Weapons would demonstrate

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17 Ms Alexandra Kelton, Head of Policy, Strategy and Engagement Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, p. 7.

18 Ms Alexandra Kelton, Head of Policy, Strategy and Engagement Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, p. 7.

19 Ms Alexandra Kelton, Head of Policy, Strategy and Engagement Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, pages 6 and 11.

20 Professor Donald Rothwell, Private Capacity, *Committee Hansard*, Canberra, 16 October 2024, p. 5.

21 Professor Donald Rothwell, Private Capacity, *Committee Hansard*, Canberra, 16 October 2024, p. 5.

22 Ms Sophia Blix, Acting Head, Stewardship and Security Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, p. 14.

23 Ms Sophia Blix, Acting Head, Stewardship and Security Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, p. 14.

24 Professor Donald Rothwell, Private Capacity, *Committee Hansard*, Canberra, 16 October 2024, p. 6.

25 Mr David Sweeney, Nuclear Policy Analyst, ACF, *Committee Hansard*, Canberra, 16 October 2024, p. 17.

that the Agreement is 'not a stepping stone' towards acquiring nuclear weapons and would be a reassurance to the Indo-Pacific.[26]

4.16 Caring for South Australia shared these views and stated in their submission that the Agreement potentially breached the Treaty on the Non-Proliferation of Nuclear Weapons (NPT) that Australia is a signatory to.[27] [28] The submission stated that Article VI(H) prohibiting the detonation of a nuclear weapon in Australia suggested the potential for there to be a nuclear weapon on Australian territory.[29] Friends of the Earth Adelaide emphasised the geopolitical importance of the NPT, stating that the Agreement would facilitate Australia acquiring highly enriched uranium and could set a precedent that resulted in other countries acquiring nuclear weapons and undermining the disarmament regime.[30]

4.17 The Australian Nuclear Association (ANA) opposed this viewpoint, stating that the Agreement did not involve the technology required for nuclear weapons.[31] They stated that the Agreement met the highest non-proliferation standards and obliged Australia to:

> …only use nuclear materials transferred under the Agreement for naval propulsion purposes, to not enrich or reprocess any material received and to be responsible for the management, disposition, storage and disposal of any spent fuel and radioactive waste resulting from naval nuclear propulsion plants.[32]

4.18 The ASA stated that the Agreement was consistent with Australia's existing non-proliferation obligations under the NPT and Treaty of Rarotonga and Australia's safeguard agreements with the International Atomic Energy Agency (IAEA).[33] The ASA also emphasised that the nuclear material in UK and US nuclear-powered submarines was not in a form that could be used in nuclear weapons without further processing by facilities that Australia does not have and will not acquire.[34] The Department of Foreign Affairs and Trade highlighted the Agreement's provision that nuclear material and equipment was unable to be transferred until Australia had concluded negotiating the Article 14 Arrangement with the IAEA.[35]

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26 Mr David Sweeney, Nuclear Policy Analyst, ACF, *Committee Hansard*, Canberra, 16 October 2024, p. 18.

27 Sister Michele Madigan, Member, Caring for South Australia, *Committee Hansard,* Adelaide, 18 October 2024, pages 1-2.

28 Caring for South Australia, *Submission 159,* pages 3-4.

29 Caring for South Australia, *Submission 159,* pages 3-4.

30 Mr Philip White, Member, Administration and Strategy Collective, Friends of the Earth Adelaide, *Committee Hansard,* Adelaide, 18 October 2024, p. 6.

31 Dr John Harries, Secretary, Australian Nuclear Association (ANA), *Committee Hansard*, Canberra, 16 October 2024, p. 3.

32 Dr John Harries, Secretary, ANA, *Committee Hansard*, Canberra, 16 October 2024, p. 1.

33 Ms Alexandra Kelton, Head of Policy, Strategy and Engagement Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, p. 6.

34 Ms Sophia Blix, Acting Head, Stewardship and Security Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, p. 7.

35 Mr Lloyd Brodrick, Acting First Assistant Secretary, International Security Division, Department of Foreign Affairs and Trade, *Committee Hansard*, Canberra, 24 October 2024, p. 11.

Committee comment

4.19 The Committee appreciates concerns raised with the wording of the Agreement but simultaneously values the strength of the relationships with the other Parties. The Committee recognises the benefit to the national interest that the AUKUS program provides and that the program cannot be delivered without ratification of the Agreement.

4.20 The Committee understands why issues relating to indemnity and other Parties withdrawing from the Agreement have been raised but believes these to be reflective of the lifespan and complexity of the Agreement rather than a treaty that is unfairly weighted against Australia. The Committee agrees that there is a high likelihood of instruments resulting from ratifying the Agreement and strongly believes that they should receive appropriate parliamentary scrutiny.

4.21 The Committee was reassured that the Australian Government remains committed to the Treaty on the Non-Proliferation of Nuclear Weapons and Treaty of Rarotonga. Additional reassurance is provided by ongoing negotiations with the IAEA relating to the Article 14 Arrangement. These commitments are reflected in the Agreement and satisfy the Committee's concerns regarding the Agreement resulting in Australia acquiring nuclear weapons.

Recommendation 3

4.22 The Committee recommends appropriate parliamentary scrutiny of any supplementary or additional instruments resulting from ratifying the AUKUS treaty.

5. Nuclear material

Site safety

5.1 Evidence provided to the Committee over the course of the inquiry highlighted public fear and concerns for safety over the presence of nuclear material at the sites across Australia being used to build and maintain nuclear submarines. This includes fears of nuclear reactor accidents,[1] submarines sinking,[2] dangers to human health,[3] and the inability of the relevant regulators to prevent safety non-compliance.[4]

5.2 In addition, the Committee received a number of submissions raising concerns about nuclear safety and the potential risks to local communities despite, as the Royal Institution of Naval Architects (RINA) told the Committee, nuclear-powered submarines being safely operated by other countries for 'well over 50 years.'[5]

5.3 Further, the witnesses discussed how transparency is essential to ensuring social license of the AUKUS sites:

> To get social license, you've got to have a very public explanation of what is intended and what the consequences and implications are for the people living there.[6]

5.4 When questioned at the public hearing in Adelaide about the need for improving community engagement and consultation to obtain nuclear social licence, ASC Pty Ltd stated that while it was doing some proactive work in this space, this role would ultimately fall to Australian Submarine Agency (ASA).[7] ASC Pty Ltd also stated that it had commenced community engagement activities around the Osborne site aimed at addressing concerns about safety and environmental management.[8]

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1 Mr Phillip White, Member, Administration and Strategy Collective, Friends of the Earth Adelaide, *Committee Hansard*, Adelaide, 18 October 2024, p. 2.

2 Mr Phillip White, Member, Administration and Strategy Collective, Friends of the Earth Adelaide, *Committee Hansard*, Adelaide, 18 October 2024, p. 2; Stop AUKUS WA, *Submission 150*, p. 2.

3 Sister Michele Madigan, Member, Caring for South Australia, *Committee Hansard*, Adelaide, 18 October 2024, p. 3.

4 Ms Mia Pepper, Campaign Director, Stop AUKUS WA, *Committee Hansard*, Perth, 22 October 2024, p. 5.

5 Mr Robin Gehling, Secretary, Australian Division, The Royal Institution of Naval Architects, *Committee Hansard*, Canberra, 24 October 2024, p. 1.

6 Dr Chris Johansen, Member, Stop AUKUS WA, *Committee Hansard*, Perth, 22 October 2024, p. 9.

7 Mr Paul Gay, Chief Operating Officer (Multi Class Sustainment), ASC Pty Ltd, *Committee Hansard,* Adelaide, 18 October 2024, pages 8-9.

8 Mr Paul Gay, Chief Operating Officer (Multi Class Sustainment), ASC Pty Ltd, *Committee Hansard,* Adelaide, 18 October 2024, pages 8-9.

5.5 At the public hearing in Canberra, representatives for the ASA stated:

> It is certainly part of our social licence strategy to include community engagement with regard to radioactive waste, including by increasing the understanding of what's really involved. We have set up community liaison teams in both South Australia and Western Australia to help start that process, and we are further developing and maturing our strategy and enhancing the engagement arrangements across those communities to build awareness of and comfort with what will be occurring at HMAS *Stirling*. With respect to the local council, I would say that there is deep and ongoing engagement with the councils, particularly in WA and South Australia, and that's a very important part of the engagement that many of my colleagues in the Australian Submarine Agency are undertaking on a regular basis. Ms Kelton has also been very involved in those consultations. So we are very much committed to building community awareness, engaging with communities about their concerns and providing information, where we can, to address those.[9]

Nuclear waste

5.6 The Agreement states that:

> Australia shall be responsible for the management, disposition, storage, and disposal of any spent nuclear fuel and radioactive waste resulting from the operation of Naval Nuclear Propulsion Plants transferred pursuant to this Article, including radioactive waste generated through submarine operations, maintenance, decommissioning, and disposal.[10]

5.7 Some submitters expressed concerns to the Committee about the amount of nuclear waste that Australia will be responsible for,[11] and the prospect of it being 'dumped' on land without community consent, especially in the case of Indigenous communities.[12]

5.8 Others were worried that Australia would be 'pressured' to accept nuclear waste from the United Kingdom (UK) or United States of America (US), however the Australian Government confirmed that this is not the case.[13] The *Australian Naval Nuclear Power Safety Act 2024* clarifies that storage or disposal in Australia of spent nuclear fuel that is not from an Australian submarine is prohibited.[14]

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9 Ms Sophia Blix, Acting Head, Stewardship and Security Division, Australian Submarine Agency (ASA), *Committee Hansard*, Canberra, 24 October 2024, pages 8-9.

10 Agreement among the Government of Australia, the Government of the United Kingdom of Great Britain and Northern Ireland, and the Government of the United States of America for Cooperation Related to Naval Nuclear Propulsion, Article IV (D).

11 Mr Phillip White, Member, Administration and Strategy Collective, Friends of the Earth Adelaide, *Committee Hansard*, Adelaide, 18 October 2024, p. 4.

12 Stop AUKUS WA, *Submission 150*, p. 4; Sister Michele Madigan, Member, Caring for South Australia, *Committee Hansard*, Adelaide, 18 October 2024, p. 3.

13 Friends of the Earth Adelaide, *Submission 20*, p. 2.

14 *Australian Naval Nuclear Power Safety Act 2024*, 8A.

5.9 Currently, the nuclear waste which is produced in Australia usually takes the form of small articles such as wipes and gloves from the medical industry and is referred to as low-level or medium-level waste.[15] Australia has not stored high-level waste, such as would be generated by nuclear-powered submarines, in the country previously.[16]

5.10 Some low-level and medium-level waste will be created during routine maintenance of the nuclear-powered submarines.[17] This is being taken into account in the design for the upgrade of HMAS *Stirling*, where it is expected it will be stored until 'such time as it may be economical or required to transfer it.'[18]

5.11 As previously stated, fuel waste from nuclear-powered submarines is regarded as high-level nuclear waste. According to the ASA, the amount of nuclear fuel in each submarine's reactor would be the equivalent in volume to a wheelie bin:[19]

> You've got eight of those, one for each reactor from the submarines that we will acquire. That is over many, many decades of course, because they won't all reach end of life at the same time.[20]

5.12 This high-level waste will require a correspondingly higher level of management and security to store.[21] Details about the storage of the waste are not laid out in the Agreement,[22] and no permanent site for nuclear waste storage has been determined in Australia.[23] The submarines will not start reaching end-of-life until the 2050s, and the radioactive material will remain contained in welded 'power units' until that time.[24]

5.13 Australia is already compliant regarding the storage of low and medium-level waste, and will no doubt continue to be compliant with high level waste storage.[25]

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15 Ms Sophia Blix, Acting Head, Stewardship and Security Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, p. 8.

16 Dr John Harries, Secretary, Australian Nuclear Association (ANA), *Committee Hansard*, Canberra, 16 October 2024, p. 2.

17 Ms Sophia Blix, Acting Head, Stewardship and Security Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, p. 8.

18 Ms Sophia Blix, Acting Head, Stewardship and Security Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, p. 8.

19 Ms Sophia Blix, Acting Head, Stewardship and Security Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, p. 8.

20 Ms Sophia Blix, Acting Head, Stewardship and Security Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, p. 8.

21 Dr John Harries, Secretary, ANA, *Committee Hansard*, Canberra, 16 October 2024, p. 2.

22 Professor Donald Rothwell, Private Capacity, *Committee Hansard*, Canberra, 16 October 2024, pages 10-11.

23 Stop AUKUS WA, *Submission 150*, p. 1.

24 Ms Sophia Blix, Acting Head, Stewardship and Security Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, p. 8.

25 Dr John Harries, Secretary, ANA, *Committee Hansard*, Canberra, 16 October 2024, p. 2; Professor Donald Rothwell, Private Capacity, *Committee Hansard*, Canberra, 16 October 2024, pages 10-11.

Australian Naval Nuclear Power Safety Regulator

5.14 On 24 October 2024 the *Australian Naval Nuclear Power Safety Bill 2024* received assent.[26] The *Australian Naval Nuclear Power Safety Act 2024* establishes a framework to regulate the nuclear safety aspects of Australia's nuclear-powered submarine enterprise, including regulating the storage and disposal of spent nuclear fuel from Australian submarines.[27]

5.15 In addition, the Act establishes the Australian Naval Nuclear Power Safety Regulator as a statutory agency located within the Defence portfolio which will operate within Australia's existing system of regulation.[28]

5.16 Throughout the course of the inquiry, several witnesses raised concerns about the independent regulator residing within the Defence portfolio, suggesting that this would likely result in less transparency.

5.17 Caring for South Australia stated that:

> …the current bill provides the minister [for Defence] to give the regulator directions in some circumstances. This is not independence, and it's a serious flaw in the proposed legislation in a democratic country.
>
> While we note that the minister is required to table in each House of Parliament a statement that such direction was given to the regulator, we note again with concern that this is merely an information exercise after the event that in no way inhibits the power of the minister before he acts.[29]

5.18 When asked by the Committee how it was ensuring that the independent regulator is independent, the ASA stated:

> …the new Australian naval nuclear safety regulator, will sit within the Defence portfolio but will be independent from the ADF [Australian Defence Force] and the broader Department of Defence. A number of measures were taken as a result of the consultations in the preparation of that bill to address some of the issues that you have raised, including ensuring that there is a 12 month period between someone finishing in the ADF and potentially taking up the head role of that agency, as well as ensuring that the head of the new regulator cannot be part of the ADF. So there are some important measures in there. Really, the reason why the regulator is sitting within the Defence portfolio is the unique circumstances

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26 Australian Naval Nuclear Power Safety Bill 2024, https://www.aph.gov.au/Parliamentary_Business/Bills_Legislation/Bills_Search_Results/Result?bId=r7104, accessed 20 November 2024.

27 *Australian Naval Nuclear Power Safety Act 2024*, Sect 14(1)(d).

28 New legislation to ensure naval nuclear power safety, https://www.minister.defence.gov.au/media-releases/2023-11-16/new-legislation-ensure-naval-nuclear-power-safety, accessed 20 November 2024.

29 Mrs Robyn Jenkin, Member, Caring for South Australia, *Committee Hansard*, Adelaide, 18 October 2024, p. 1.

required for the regulation of nuclear-powered submarines and the specific military context in which they operate.[30]

5.19 Questioned separately, RINA stated that although they had initial concerns about the 'regulator becoming another branch of Defence' they believe 'the amendments accepted by the government in that bill, when it was passed, go some way towards securing that independence.'[31]

Committee comment

5.20 The Committee acknowledges that some people in the community have fears around nuclear energy and radioactive waste. The Australian nuclear industry is small, and is contained mainly to medicine and research. A high profile, high level nuclear defence project such as AUKUS is bound to raise concerns in a country where the population does not have much familiarity or experience with nuclear energy.

5.21 The Committee is satisfied that the evidence presented by the Department of Defence, ASA and relevant departments and agencies disproved many of the common AUKUS 'myths'.

5.22 It is for this reason that the Committee believes that the public must be kept truly engaged with the development of the AUKUS program, especially in the areas where nuclear material will be used, transported, or stored, to support the social licence of the navy sites to operate.

5.23 The Committee considers it of particular importance that the Australian Government ensure that its development of a nuclear waste plan for the submarines, reactors and any other related radioactive waste be as transparent as possible, and engage directly with affected communities in order to alleviate any safety concerns or misinformation.

5.24 The Committee was pleased to hear that some concerns surrounding the location of the Australian Naval Nuclear Power Safety Regulator within the Defence portfolio have been addressed, and that the regulator would remain independent from the ADF and the broader Department of Defence.

5.25 The Committee accepts some opponents to AUKUS will never be satisfied as their objection is an ideological opposition to nuclear energy. In a robust democracy such as Australia, this inquiry has allowed throughout the submissions and hearings process such voices to be heard.

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30 Ms Sophia Blix, Acting Head, Stewardship and Security Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, p. 7.

31 Mr Robin Gehling, Secretary, Australian Division, RINA, *Committee Hansard*, Canberra, 24 October 2024, p. 3.

Recommendation 4

5.26 The Committee recommends that community and worker consultation, engagement and public transparency be prioritised in the development of the plan for the management, transportation and storage of AUKUS-related nuclear waste.

Recommendation 5

5.27 The Committee recommends that advice published by the Australian Naval Nuclear Power Safety Regulator be made publicly available within a timely fashion to enhance transparency.

6. Employment and education impacts

Skills shortages

6.1 A number of submitters to the inquiry raised concerns about the potential shortage in the Australian workforce and skills required to implement the AUKUS program. One submitter, The Australia Institute, raised this issue and stated that Australia did not currently possess the knowledge to engineer the technology required by the nuclear-powered submarines.[1] Mr Allan Behm stated that this current knowledge gap would have an impact upon Australia's capability sovereignty as only officers within the United States of America (US) Navy would be fully familiar with the operations of the submarines.[2]

6.2 Mr Behm suggested that this skills shortage could be addressed but stated that there would be 'huge' costs to develop the professional, technical and operational expertise required for the construction and operation of the submarines.[3] He also highlighted that there were already existing pressures on the manufacturing and construction sectors and that any policies to address skills shortages could not be made in isolation from these pressures.[4]

6.3 Similarly, the Royal Institution of Naval Architects (RINA) stated that while supportive of the Agreement, they agreed there was currently a skills gap.[5] At the second public hearing in Canberra, Mr Robin Gehling said that one of the two Australian education institutions that offered naval architecture courses currently enrolled fewer than 20 students per annum and that the AUKUS naval shipbuilding program would require approximately 100 per year.[6] He further emphasised that the complexity in designing, constructing and operating the submarines and the lengthy timeframe, potential loss

1 Mr Allan Behm, Director, International and Security Affairs, The Australia Institute, *Committee Hansard*, Canberra, 16 October 2024, p. 11.
2 Mr Allan Behm, Director, International and Security Affairs, The Australia Institute, *Committee Hansard*, Canberra, 16 October 2024, pages 11-12.
3 Mr Allan Behm, Director, International and Security Affairs, The Australia Institute, *Committee Hansard*, Canberra, 16 October 2024, p. 13.
4 Mr Allan Behm, Director, International and Security Affairs, The Australia Institute, *Committee Hansard*, Canberra, 16 October 2024, pages 13-14.
5 Mr Robin Gehling, Secretary, Australian Division, Royal Institution of Naval Architects (RINA), *Committee Hansard,* Canberra, 24 October 2024, p. 1.
6 Mr Robin Gehling, Secretary, Australian Division, RINA, *Committee Hansard,* Canberra, 24 October 2024, p. 4.

of expertise, workforce recruitment and workforce management must be 'taken into account to ensure safe and reliable outcomes'.[7]

6.4 When asked by the Committee about the current skills gap in Australia, the Australian Nuclear Association (ANA) agreed that while there was currently a shortage of workers skilled in areas required, they believed that the staged design of the Agreement gave Australia enough time for the shortage to be addressed.[8]

6.5 The Australian Submarine Agency (ASA) noted that they were seeking to address this gap by working with the tertiary and vocational education sectors to create incentives and upskill the Australian workforce, citing the announcement of 4,001 Commonwealth-supported STEM placements as a specific initiative to address the skills shortage.[9] The ASA also highlighted ongoing training programs to upskill Royal Australian Navy personnel and ensure their ability to operate nuclear-powered submarines and related technologies.[10]

6.6 When asked by the Committee about the forecast number of submariners required by the Australian Navy to operate both the Virgina class submarines and the Australian built SSN-AUKUS submarines, the Department of Defence stated that they would require a larger contingent of submariners in the Navy to meet the AUKUS program's needs.[11] Commodore Michael Jacobson said that the Navy would need to increase the figure of 900 submariners to 3,000 by the early 2050s, but that the length of the program afforded them the time necessary to do so and that the Navy was achieving the growth targets required to meet their capability requirements.[12]

Employment and education benefits

6.7 The employment and education benefits of the AUKUS program were identified in evidence presented to the Committee. ASC Pty Ltd, who are jointly responsible for building the nuclear-powered submarines, said that the Agreement would create jobs and supply chain opportunities.[13] They stated that they were currently recruiting, were

7 Mr Robin Gehling, Secretary, Australian Division, RINA, *Committee Hansard,* Canberra, 24 October 2024, p. 2.

8 Dr John Harris, Secretary, Australian Nuclear Association (ANA), *Committee Hansard*, Canberra, 16 October 2024, p. 4.

9 Ms Alexandra Kelton, Head of Policy, Strategy and Engagement Division, Australian Submarine Agency (ASA), *Committee Hansard*, Canberra, 24 October 2024, p. 10.

10 Ms Alexandra Kelton, Head of Policy, Strategy and Engagement Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, p. 12.

11 Commodore Michael Jacobson, Director General, Submarines, Department of Defence, *Committee Hansard*, Canberra, 24 October 2024, p. 12.

12 Commodore Michael Jacobson, Director General, Submarines, Department of Defence, *Committee Hansard*, Canberra, 24 October 2024, p. 12.

13 Mr Alex Walsh, Chief Nuclear and Capability Officer, ASC Pty Ltd, *Committee Hansard*, Adelaide, 18 October 2024, p. 8.

creating a workforce plan for ongoing recruitment and skills uplift, and had already launched large graduate and apprentice recruitment programs.[14] [15]

6.8 The ASA described the AUKUS program to the Committee as 'an unparalleled opportunity for our nation', stating that the program would create approximately 20,000 jobs over 30 years across industry, the Australian Defence Force and Australian Public Service.[16] The ASA stated that the total number of jobs included the 4,000 required to design and build the infrastructure for the submarine construction, 6,000 to support the submarine build, 3,000 to support submarine activities at Henderson and 3,000 to upgrade HMAS *Stirling* in support of AUKUS.[17] The ASA further stated that the long length of the Agreement would result in many of the jobs created by the AUKUS program lasting 'well into the 2050s'.[18]

6.9 The variety of the jobs created by the AUKUS program was touted, with ASA stating that jobs created would include lawyers, policymakers, nuclear experts, high-tech STEM roles, project managers, financial experts and manufacturers.[19]

6.10 In answers to questions on notice from the Committee, the ASA expanded on the education benefits provided for by the AUKUS program. In addition to the 4,001 Commonwealth-supported STEM placements, the Australian Government has announced $68.4 million for initiatives delivered through a new Skills and Training Academy in South Australia and $16.3 million to establish the Nuclear-Powered Submarine Student Support Scholarships Program.[20] The program is expected to deliver 3,000 scholarships for students studying undergraduate STEM courses relevant to the nuclear-powered submarine enterprise.[21]

6.11 The ASA detailed ongoing programs that support Australians from the nuclear-powered submarine enterprise in undertaking undergraduate and postgraduate nuclear studies.[22] They also highlighted placements at Pearl Harbor Naval Shipyard in the US for ASC Pty Ltd personnel undertaking training relating to nuclear and non-nuclear skills.[23]

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14 Mr Paul Gay, Chief Operating Officer (Multi Class Sustainment) ASC Pty Ltd, *Committee Hansard*, Adelaide, 18 October 2024, p. 9.

15 Mr Alex Walsh, Chief Nuclear and Capability Officer, ASC Pty Ltd, *Committee Hansard*, Adelaide, 18 October 2024, p. 10.

16 Ms Alexandra Kelton, Head of Policy, Strategy and Engagement Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, p. 5.

17 Ms Alexandra Kelton, Head of Policy, Strategy and Engagement Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, pages 5-6.

18 Ms Alexandra Kelton, Head of Policy, Strategy and Engagement Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, p. 9.

19 Ms Alexandra Kelton, Head of Policy, Strategy and Engagement Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, p. 10.

20 Answers to questions on notice provided by the ASA at a public hearing in Canberra, 24 October 2024, p. 2.

21 Answers to questions on notice provided by the ASA at a public hearing in Canberra, 24 October 2024, p. 2.

22 Answers to questions on notice provided by the ASA at a public hearing in Canberra, 24 October 2024, p. 2.

23 Answers to questions on notice provided by the ASA at a public hearing in Canberra, 24 October 2024, p. 2.

Committee comment

6.12 The Committee believes the potential shortage of skills required to implement the AUKUS program is a significant issue but was encouraged to hear about the initiatives already being undertaken to address the shortages. The Committee is satisfied that the relevant areas of government are cognisant of the issue and that it can be addressed through continued hiring and training initiatives from government and private industry.

6.13 The Committee believes that by Australia working with AUKUS partners to build the pathways for international placements, the Agreement presents a unique opportunity to upskill Australia's domestic industrial workforce. Australia will retain and benefit from this upskilled workforce for the life of the Agreement and beyond.

6.14 The jobs created throughout the life of the AUKUS program is good news to the Committee, as is the longevity of the employment opportunities that will be created. The Committee sees the skills shortage as an opportunity for a diversified workforce and the growth in employment to be in Australia's interests.

6.15 From the evidence provided throughout this inquiry, the Committee agrees the Agreement is a whole-of-nation undertaking that will see around 20,000 secure high paying jobs created across Australia, with unprecedented opportunities for trades and tertiary educated workers alike.

Recommendation 6

6.16 The Committee recommends that the Australian Government launch an AUKUS career and employment opportunities promotion campaign and further enhance existing STEM and traditional trades engagement programs for local schools and tertiary education providers.

7. Local community impact

AUKUS sites

7.1 The Agreement does not stipulate the location of where the nuclear-powered submarines will be built or where the Virgina class submarines will be stored, maintained or operated. However, evidence provided by the Australian Submarine Agency (ASA) identified Osborne Shipyard in South Australia as the construction site for the submarines, and HMAS *Stirling* and Henderson in Western Australia as the locations where the Virgina class submarines will be maintained.[1]

7.2 At the Osborne Naval Shipyard, a construction site is being developed specifically for the building of the SSN-AUKUS submarines. Australian Naval Infrastructure (ANI), a Government Business Enterprise whose primary objective is to 'support the Commonwealth's continuous naval shipbuilding program,'[2] and the ASA, which was established to 'acquire, construct, deliver, technically govern, sustain and dispose of' AUKUS submarines, are partnering in this project.[3]

7.3 Henderson WA was announced in 2024 as the site of a new defence precinct which will include a 'depot-level maintenance and contingency docking' for nuclear-powered submarines.[4]

7.4 HMAS *Stirling* provides 'operational and logistical support to Royal Australian Navy ships, submarines and aircraft based in WA.'[5] This site will have a rotational presence of AUKUS partner submarines from 2027, which will be known as the Submarine Rotational Force – West.[6] The first submarines to operate out of this base will be Virginia class submarines from the United States of America (US) and an Astute class submarine from the United Kingdom (UK).[7]

7.5 The required upgrades and ongoing increase in personnel at all three locations have an impact on local communities within the areas of the Osborne in South Australia,

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1 Ms Alexandra Kelton, Head of Policy, Strategy and Engagement Division, Australian Submarine Agency (ASA), *Committee Hansard*, Canberra, 24 October 2024, pages 5-6.

2 About ANI, www.ani.com.au/about, accessed 14 November 2024.

3 About, www.asa.gov.au/about, accessed 14 November 2024.

4 New Defence Precinct at Henderson to deliver continuous Naval Shipbuilding and Nuclear-Powered Submarine Maintenance, www.asa.gov.au/news/new-defence-precinct-henderson-deliver-continuous-naval-shipbuilding-and-nuclear-powered-submarine-maintenance, accessed 14 November 2024.

5 HMAS *Stirling*, www.navy.gov.au/about-navy/bases-and-locations/hmas-stirling, accessed 14 November 2024.

6 Submarine Rotational Force – West Infrastructure Project, www.asa.gov.au/projects/submarine-rotational-force-west-infrastructure-project, accessed 14 November 2024.

7 Ms Alexandra Kelton, Head of Policy, Strategy and Engagement Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, p. 6.

and HMAS *Stirling* and Henderson in Western Australia. These impacts will mostly result from an increase in traffic coming and going from the above-mentioned sites, increase to the population living and working in the local areas, and the associated building works required within the sites.

Infrastructure impacts

7.6 Concerns with the impact of implementing the AUKUS program would have upon local infrastructure was raised in public hearings. The Conservation Council of Western Australia referenced that the program's development in and around Henderson and HMAS *Stirling* and said that it would lead to increased stress upon local roads and waters.[8]

7.7 As part of the inquiry the Committee travelled to Osborne in South Australia and HMAS *Stirling* and Henderson in Western Australia. During its visit, the Committee was made aware of the current and future infrastructure concerns to the local communities as a result of the AUKUS program.

7.8 The Conservation Council of Western Australia said that the increased number of workers required by the AUKUS program would result in additional people seeking housing in an area that already has a housing crisis and would place a strain on already strained services.[9] Stop AUKUS WA added to these concerns, stating that recent housing announcements for Australian Defence Force housing raised questions as to why housing was not being built for local community members.[10]

7.9 When questioned about whose responsibility it was to mitigate these infrastructure impacts resulting from increased populations to the neighbouring local communities, ASA stated:

> …the Australian Submarine Agency and Defence have been working very closely with both the WA State Government and Local governments to understand the impact of, as you say, the increased levels of activity and what that will mean for the local community. We have held in recent months two meetings that we call three-tier-of-government meetings—that is, federal ministers including Minister King, state level WA ministers and local councils—to unpack and understand these issues.[11]

7.10 Following the hearing the Department of Defence (Defence) provided additional information:

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8 Ms Mia Pepper, Campaign Director, Conservation Council of Western Australia, *Committee Hansard*, Perth, 22 October 2024, Canberra, p. 4.

9 Ms Mia Pepper, Campaign Director, Conservation Council of Western Australia, *Committee Hansard*, Perth, 22 October 2024, Canberra, p. 4.

10 Dr Chris Johansen, Member, Stop AUKUS WA, *Committee Hansard*, Perth, 22 October 2024, p. 4.

11 Ms Alexandra Kelton, Head of Policy, Strategy and Engagement Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, p. 9.

> Defence is working closely with the Western Australian Government and relevant local councils, to ensure Defence needs at HMAS *Stirling* and Henderson align with the needs of the Western Australian Government, local councils and industry, and neighbouring communities.
>
> At Osborne, the Australian Submarine Agency and Australian Naval Infrastructure, the Government's appointed infrastructure delivery partner, are working collaboratively with the South Australian Government to progress master-planning for the Lefevre Peninsula. This will ensure the LeFevre Peninsula is well placed to deliver on its critical national role, whilst also supporting industry, and the local community. The South Australian Government has recently appointed COX Pty Ltd to undertake master planning for the Lefevre Peninsula. This updated master plan will support a range of activities, including traffic flow, emergency services and community spaces.[12]

Engagement with First Nations community

7.11 Although no witnesses nor submitters who engaged with the inquiry identified themselves to the Committee as First Nations or representatives of Traditional Owner groups, throughout the inquiry several witnesses did raise concerns about the lack of consultation with First Nations people.

7.12 Particular concern was raised in regard to the AUKUS program's ongoing compliance with the United Nations Declaration on the Rights of Indigenous Peoples (UNDRIP), and the lack of consultation with First Nations people to whom Garden Island in Western Australia holds a level of significance.[13]

7.13 While no evidence was presented to this Committee regarding engagement conducted with First Nations people or Traditional Owner groups, the Parliamentary Standing Committee on Public Works was advised by Defence that as part of the *Environment Protection and Biodiversity Conservation Act 1999* referral process for *Submarine Rotational Force—West, Priority Works*, HMAS *Stirling*, Western Australia it had consulted with:

- Southwest Aboriginal Land and Sea Council;
- Aboriginal Advisory Council of Western Australia; and
- Gnaala Karla Booja Aboriginal Corporation.[14]

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12 Answers to questions on notice provided by the Department of Defence (08) at a public hearing in Canberra, 24 October 2024, p. 1.

13 Ms Leonie Lundy, Campaign Coordinator, Stop AUKUS WA, *Committee Hansard*, Perth, 22 October 2024, p. 9.

13 Ms Leonie Lundy, Campaign Coordinator, Stop AUKUS WA, *Committee Hansard*, Perth, 22 October 2024, p. 9.

14 Department of Defence*, Submission 1 to the* Parliamentary Standing Committee on Public Works inquiry into Submarine Rotational Force—West, Priority Works, HMAS *Stirling*, Western Australia, June 2024, p. 15.

Environmental impacts

7.14 At the public hearing in Perth, the Conservation Council of Western Australia outlined a number of environmental impacts which may be realised through the implementation of the AUKUS program:

> There will be housing requirements for any personnel both on the island and onshore. There will be increasing human traffic on the island, so that will impact the environmental communities that exist there. Some of the development proposals for Cockburn Sound—for example, dredging to support submarines coming in and also upgrading the docking areas—will interfere with nesting areas for little penguins and foraging grounds. Seagrass meadows will be dredged as part of the proposal. There will be upgrades to the causeway. There will presumably be upgrades to roads servicing the area, and these will all be in areas where there are threatened ecological communities, which will be further fragmented through that area.[15]

7.15 When asked by the Committee what environmental considerations were being made to ensure there were no or minimal environmental impacts to the local areas Defence stated:

> All of Defence's infrastructure works are required to comply with the requirements of the *Environment Protection and Biodiversity Conservation Act 1999* (EPBC Act). Defence and the Australian Submarine Agency have referred major construction at both Osborne and HMAS *Stirling* for consideration by the Minister for the Environment and Water, the Hon Tanya Plibersek MP, under the EPBC Act.[16]

7.16 The Conservation Council of Western Australia raised concerns that the platform for considering environmental impacts is 'not right' as the commitment to the Agreement and locations of the AUKUS sites had already been announced prior to any environmental assessment becoming undertaken:[17]

> They've already named the location, both of HMAS *Stirling* and Henderson. It feels like, if we were to input into that process, it's not as if the DCCEEW [Department of Climate Change, Energy, the Environment and Water] or the EPA [Environment Protection Authority] would then be in a position to say, 'Actually, this site is an inappropriate location for this development based on environmental grounds and should not go ahead.'[18]

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15 Dr Kelly Duckworth, Policy and Research Manager, Conservation Council of Western Australia, *Committee Hansard*, Perth, 22 October 2024, p. 6.

16 Answers to questions on notice provided by the Department of Defence (09) at a public hearing in Canberra, 24 October 2024, p 1.

17 Ms Mia Pepper, Campaign Director, Conservation Council of Western Australia, *Committee Hansard*, Perth, 22 October 2024, pages 4 and 8.

18 Ms Mia Pepper, Campaign Director, Conservation Council of Western Australia, *Committee Hansard*, Perth, 22 October 2024, p. 8.

7.17 In response Defence told the Committee that:

> The Department of Defence is committed to working closely with the Department of the Environment, Climate Change and Water to do all the processes in line with getting the requisite approvals and consultations.[19]

Committee comment

7.18 The AUKUS program's impact upon local infrastructure was of concern to the Committee. To address these concerns fully, the Committee believes both ASA and the Department of Defence should consult with the local communities, publicise any planning initiatives to enable substantive engagement from locals, and provide publicly accessible planning documents to help build social license for infrastructure initiatives going forward.

7.19 The Committee recognises that the increased workforce and activity at each of the AUKUS sites will have substantial impacts on community infrastructure, the cost of which should not be shouldered by the local governments and communities alone. The Committee acknowledges these impacts and advocates for funding to be made available to Local and State governments to address these issues as they arise.

7.20 While the Committee acknowledges the consultation Defence has undertaken to date with First Nations people and Traditional Owner groups regarding the works at HMAS *Stirling*, the Committee believes that a greater level of ongoing consultation is required to ensure that First Nations concerns are addressed across all locations and aspects of the AUKUS program.

7.21 The Committee acknowledges the requirement for any proposed works that may significantly impact to be referred to the environment to the Minister for the Environment and Water under the EPBC Act, however, notes that some ecological impacts resulting from the AUKUS program, such as the impacts resulting from the increase to local populations may fall outside the EPBC Act and Australian Government jurisdiction.

7.22 The Committee recognises the appearance of planning approvals as "piecemeal", however notes that the multi part process reflects the complexity of cross-jurisdictional planning approval processes in Australia.

Recommendation 7

7.23 The Committee recommends that the Government partner with relevant State and Local governments to collaborate and coordinate on planning and delivery of enabling programs, infrastructure and works to support communities involved in the implementation of AUKUS.

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[19] Ms Alexandra Kelton, Head of Policy, Strategy and Engagement Division, ASA, *Committee Hansard*, Canberra, 24 October 2024, p. 9.

Recommendation 8

7.24 The Committee recommends that the Australian Submarine Agency enhance engagement and consultation with the local Traditional Owner group and First Nations peoples; and with local environmental groups.

8. Conclusion

8.1 The Joint Standing Committee on Treaties was established by the Commonwealth Parliament to inquire into any treaty to which Australia has become signatory. It is the role of the Committee to consider both the treaty and matters arising from it in order to ensure that ratification is in the national interest.

8.2 As part of its inquiry into the *Agreement among the Government of Australia, the Government of the United Kingdom of Great Britain and Northern Ireland, and the Government of the United States of America for Cooperation Related to Naval Nuclear Propulsion* (the Agreement), the Committee considered a range of matters brought to it throughout the inquiry process.

8.3 Throughout the inquiry the Committee heard evidence that, unlike diesel submarines, nuclear-powered submarines have the ability to operate at high speeds for extended periods of time without the need to surface. This ability, in addition to superior manoeuvrability and stealth, provides Australia with a clear advantage in a time when Australia is facing the most complex and challenging strategic environment since the Second World War.

8.4 While supporting Australia's ratification of the treaty, the Committee makes eight recommendations that seek to ensure that the Australian people are continuously engaged and informed throughout the implementation of the Agreement and the resulting AUKUS program.

8.5 Key among the matters these recommendations address is the need to support local communities who will be impacted by the significant increase in population brought about by the influx of workers and their families to the locations where the submarines will be built and maintained. This support should take the form of education, consultation and funding provided by the Federal Government and implemented in partnership with relevant State and Local governments.

8.6 The issue of nuclear waste disposal needs to be addressed in a timely and transparent manner. While the Committee acknowledges that the disposal of the nuclear fuel used to power the submarines will not be required until the 2050s or later, the decision of where to store the small amounts of low-level waste created during routine maintenance of the nuclear-powered submarines needs to be a priority. As part of this process, the community needs to better engaged and educated in the risks posed through the storage of the waste, and the advice provided by the newly-established Australian Naval Nuclear Power Safety Regulator should be made available to the general public to maintain transparency.

8.7 The Committee acknowledges that this treaty intersects with a range of other international agreements to which Australia is a party, or may be party to in the

future. Additionally, the Committee recognises that a number of additional instruments may arise from the ratification of the Agreement and strongly believes that they should receive appropriate parliamentary scrutiny.

8.8 Finally, the Committee was pleased to hear that the implementation of the Agreement brings with it a significant opportunity by way of generating thousands of jobs and forms the catalyst for increased focus on STEM subjects at schools and in tertiary education. The resulting upskilled workforce will benefit Australia not only throughout the life of the Agreement but well into the future.

8.9 In conclusion, the Committee believes that while the nuclear-powered submarines themselves provide Australia with a strategic advantage, the resulting upskilling of the Australian workforce, increased employment opportunities and supply chain opportunities brought about by the Agreement are of great benefit to Australia and thus the Committee recommends that binding treaty action be taken.

Ms Lisa Chesters MP

Chair

25 November 2024

A. Submissions

1	Mr Crispin Rovere
2	Mr Robert Heron
3	Ms Susan Benham
4	Mr Bevan Ramsden
5	Friends of the Earth Adelaide
6	Women's International League for Peace and Freedom Australia
7	Mr John Tonkin
8	Mr Andrew Gaines
9	Ms Amelia King
10	Dr Marty Branagan
11	Mr Tom Marwick
12	Mr Joseph Philippa
13	Mr G Toll
14	Ms Helen Breed
15	Mr Jim Hazzard
16	Mr Robert Coney
17	Ms Jennifer Lyons
18	Mr Ben Spina
19	Mr James Lechte
20	Mr Stephen Clendinnen
21	Mr Terry Barridge
22	Mr Noel Wauchope

23 Mr Michael Murphy

24 Mr John Glazebrook

25 Mr James Arnold

26 Mr Tom Curnow

27 Mr Steve Gill

28 Mr Johnston Yap

29 Mr Paul Gregory Morgan

30 Mr Farhad Billimoria

31 Ms Emma Lees

32 Mr Rieuwen Smit

33 Mr Alonso Samillan

34 Mr Stephen Due

35 Mr Victor Grech

36 Mr Robert Marotta

37 Mr Almerindo Migliore

38 Ms Bella Swan

39 Mr Emmanuelle Missios

40 Mr Richard Statham

41 Ms Paula Hall

42 Mr Colin Fraser

43 Mr John Tindall

44 Ms Wendy Balboni

45 Mr Ross Brown

46 Mr John Shaban

47 Mr Miroslav Capko

48 Dr Doug Buchanan

49 Mr Brad Degney

50 Mr Donald Jeffery Sayers

51 *Name Withheld*

52 *Name Withheld*

53 *Name Withheld*

54 *Name Withheld*

55 *Name Withheld*

56 *Name Withheld*

57 *Name Withheld*

58 *Name Withheld*

59 Mr Jeff White

60 Ms Simone Braund

61 Mr Gary Janke

62 Mr Joshua Camm

63 Mr Tom Oakley

64 Evolved

65 Mr Russell Butler

66 Mr John Shaw

67 Dr Bruce Adams

68 Ms Suzanne Vasanji

69 Mr Michael Barnes

70 Mr Frank Fisher

71 Mr Philip James

72 Mr Kenneth Wilkinson

73 Mr Wilfred Lazar

74 Ms Judy and Mr Alan Clark

75 Ms Kay Deaves

76 Ms Patricia Laubinger

77 Mr Colin Ely

78 Mr Bruce George

79 Ms Sahra Miller

80 Mr Chris Wright

81 Mrs Ann Lawler

82 Ms Elizabeth Stephens

83 Mr George Mercier

84 Ms Catherine Reely

85 Mr Warren Brown

86 Mr Jacob Flentri

87 Mr Trilokesh Chanmugam

88 Captain RAN (rtd) Christopher Skinner

89 Mr Russell Morse

90 Mr Jonathan White

91 Mr Michael Randall

92 Ms Erica Henshall

93 Ms Raylee Barrett

94 Mr Ordan Andreevski

95 Ms Marguerite Hann-Syme

96 Ms Katherine Eagles

97 Mr Ian Lloyd

98 Ms Sherry Eastwood

99 Mr Adrian Pagett

100 Ms Anne Warwick

101 Mr Jeff Lormans

102 Ms Ingrid Pich

103 Ms Lois Van Geloven

104 Mr Barry Cameron

105 Mr Robert Jameson

106 Mr Appollo Yianni

107 Mr Rod McKenzie

108 Mr Max Goulter

109 Mr David Newton

110 Ms Annette Brownlie

111 Australian Nuclear Association

112 Ms Michele Cochrane

113 Mr Peter Cowell

114 Ms Judith Doherty

115 Mr Dennis Pukallus

116 Mrs Narelle Clarke

117 Mr John Christopher Kohl

118 Ms Vicki Spencer

119 Mr Graeme Batterbury

120 Ms Nicole Everett

121 Ms Catherine Brown

122 Mr Palani Sg

123 SS Singh

124 Mr Galen Armfield

125 Mr Desmond Whyte

126 Mr Hugh Drum

127 Ms Monica Mesch

128 Ms Mary Galic

129 Ms Pamela Collett

130 Mr Ross Pearce

131 Mr Warren Ford

132 Dr Christian Sims

133 Ms Kerry Greaves

134 Mr Darko Markovic

135 Mr Aran Levin

136 Ms Nik Dangar

137 Labor Against War

138 Professor Donald Rothwell

139 Ms Kelly Paull

140 Mrs Christina Emblem

141 Ms Sarah Isaacs

142 Mr Peter Huskisson

143 Mr Rick Grange

144 Mr Kheng Ee Sturm

145 Ms Adrienne Shilling

146 Ms Judy Sudholz

147 Mr Mark McKinley

148 Mr Roger Fairfax

149 Mr James Langley

150 Stop AUKUS WA

151 Mr Justin Tutty

152 Ms Wendy Wales

153 Mr Paul Derons

154 Mr David Noonan

155 Mr Clinton Duncan

156 Ms Winnie Mak

157 Ms Jane Kung

158 Mr John Carver

159 Caring for South Australia

160 Mr Alan Kung

161 Ms Jo Lindo

162 Mr Leigh Rose

163 Professor Helen Ware

164 Mr Agnidev Ananda

165 Ms Julie Marlow

166 Mr Geoff Leury

167 Independent and Peaceful Australia Network (IPAN)

168 Just Peace Queensland

169 Dr Ross Gwyther

170 Mr Rob Gulley

171 Ms Alison Harwood

172 Ms Carolyn Chaston

173 Medical Association for Prevention of War

174 Dr Lilliana Corredor

175 Ms Kristin Hansen

176 Mr John Clark

177 Ms Peta Terry

178 Mr Frankie Seymour

179 Ms Louise Ludlam-Taylor

180 Mr Kheng Selling

181 Australian Conservation Foundation

182 Mr Brock Maurer

183 Alice Springs Peace Action Think Tank

184 Mr Kheng see Teoh

185 Mr David Schwartz

186 Ms Gayle Davies

187 Nuclear Free WA

188 Friends of the Earth Australia

189 Top End Peace Alliance

190 International Campaign for the Abolition of Nuclear Weapons Australia

191 Australians for War Powers Reform

192 Ms Helen Banks

193 Mr Sean Griffiths

194 Mr David Gray

195 Ms Susan Bollard

196 Ms Adara Thomson

197 Mr Mehdi Zahedpur

198 Ms Leonie Chester

199 Mr Leslie Farkash

200 Mr Steven Tirtapranawa

201 Ms Eila Onfray

202 Ms Helge Gram

203 Mr Terrence Shelton

204 Mr Benson Lee

205 Mr David Taylor

206 Dal MC

207 Mr Maggoo Fox

208 Mr Grant Mellon

209 Mr Peter Roche

210 Mr Nick Pastalatzis

211 Mr Jeff Leddin

212 Mr Gary Lord

213 Mr Helmut Brodesser

214 Ms Hilary Myers

215 Ms Colette Leber

216 Mr Tai Olde

217 Mr David Duckering

218 Mr Colin Jones

219 General Sir Colin Dunn VOAV

220 Mr Wayne Thompson

221 Dr David Miller

222 Mr Usama Zafar

223 Mr Stephen Lordan

224 Mr Nicholas O'Dwyer

225 Mr Roger Blakeway

226 Mr Alan Smith

227 Ms Melissa X

228 Ms Latoya Voogt

229 Ms Ann Pearse

230 The Australia Institute

231 Mr Navid Paya

232 Mr Derek Tomaseti

233 Dr Colin McQueen

234 Mr Alan Cummine

235 Ms Carin Cinnadaio

236 Confidential

237 *Name Withheld*

238 *Name Withheld*

239 *Name Withheld*

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251 *Name Withheld*

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253 *Name Withheld*

254 *Name Withheld*

255 *Name Withheld*

256 *Name Withheld*

257 *Name Withheld*

258 *Name Withheld*

259 *Name Withheld*

260 Confidential

B. Witnesses

Agreement among the Government of Australia, the Government of the United Kingdom of Great Britain and Northern Ireland, and the Government of the United States of America for Cooperation Related to Naval Nuclear Propulsion

Wednesday 16 October 2024

Committee Room 1R4, Parliament House, Canberra

Australian Nuclear Association

- Dr John Harries, Secretary

Professor Donald Rothwell, Private Capacity

The Australia Institute

- Mr Allan Behm, Director, International and Security Affairs

Australian Conservation Foundation

- Mr David Sweeney, Nuclear Policy Analyst
- Mr Adam Beeson, General Counsel

Friday 18 October 2024

Vault 2, Adina Apartment Hotel Adelaide Treasury, 2 Flinders Street, Adelaide

Friends of the Earth Adelaide

- Mr Philip White, Member of Administration and Strategy Collective

Caring for South Australia

- Mrs Robyn Jenkin, Member
- Sister Michele Madigan, Member

ASC Pty Ltd

- Mr Paul Gay, Chief Operating Officer (Multi Class Sustainment)
- Mr Alex Walsh, Chief Nuclear and Capability Officer

Tuesday 22 October 2024

Anglesea #1 Room, Mercure Perth, 10 Irwin Street, Perth

Nuclear Free Western Australia

- Ms Josephine Vallentine, Member and Patron
- Ms June Lowe, Committee Member

Stop AUKUS WA

- Dr Chris Johansen, Member
- Ms Leonie Lundy, Campaign Coordinator

Conservation Council of Western Australia

- Dr Kelly Duckworth, Policy and Research Manager
- Ms Mia Pepper, Campaign Director

Thursday 24 October 2024

Committee Room 1R4, Parliament House, Canberra

The Royal Institution of Naval Architects

- Mr Robin Gehling, Secretary, Australian Division

Department of Foreign Affairs and Trade

- Mr Lloyd Brodrick, Acting First Assistant Secretary, International Security Division
- Ms Emily Hill, Assistant Secretary, AUKUS and Partnerships Branch
- Dr Kate Mitchell, Acting Assistant Secretary, International Law Branch
- Dr Yasmin Naqvi, Director, AUKUS Trilateral Legal Frameworks Section, Legal Division

Australian Submarine Agency

- Ms Sophia Blix, Acting Head, Stewardship and Security Division
- Mr Andrew Evers, General Counsel
- Ms Alexandra Kelton, Head of Policy, Strategy and Engagement Division
- Mr Lindsay Gordon, Director, Nuclear Governance Systems

Australian Safeguards and Non-Proliferation Office

- Ms Charlotte East, Assistant Secretary, Nuclear Safeguards and Security
- Dr Craig Everton, Assistant Secretary, Non-Proliferation, Technology and Partnerships

- Mr Alistair Knight, Director, Nuclear Cooperation and Design Section, Non-Proliferation, Technology and Partnerships Branch

Attorney-General's Department

- Mr Ben Huntley, Acting Assistant Secretary, Office of International Law

Department of Defence

- Commodore Michael Jacobson, Director General, Submarines